Function Spaces: Domains, Orbit Symmetry, and Changing Numbers of Solutions

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The same function may be studied as a formula, as a map between topological spaces, or as an object carrying symmetry. A denominator determines where its formula makes sense; the image records attainable values; a functional equation can transport information across infinitely many inputs; a parameter can change the topology of a solution set. The following examples make these perspectives meet in actual calculations.

§1.1 Domains from Constraints

For the real-valued expression

$$F(x) = \frac{\sqrt{x(3-x)}}{\log_{10}((x-3)^2)},$$

the square root requires $0 \le x \le 3$, the logarithm requires $x \neq 3$, and its value cannot be zero, excluding $(x-3)^2 = 1$. Hence

$$\operatorname{dom} F = [0,2) \cup (2,3).$$

The open and closed endpoints encode different facts: $x = 0$ is allowed, while $x = 2$ gives a zero denominator and $x = 3$ gives an undefined logarithm.

A floor expression changes its form at integer thresholds:

$$\lfloor x \rfloor = k \quad \Longleftrightarrow \quad k \le x < k+1.$$

It is therefore often best handled on intervals with fixed k. The discontinuities of the floor function are a geometric record of that case split.

§1.2 Ranges and Compactness

Consider

$$G(x) = \sqrt{x-4} + \sqrt{15-3x}, \qquad 4 \le x \le 5.$$

With $u = x - 4$, this becomes

$$G = \sqrt{u} + \sqrt{3(1-u)}, \qquad 0 \le u \le 1.$$

Cauchy–Schwarz gives $G^2 \le (1+3)(u+1-u) = 4$, and equality holds at $u = 1/4$, where $G = 2$. The only interior critical point is this maximum; the endpoint values are $G(0) = \sqrt{3}$, $G(1) = 1$. Thus

$$\boxed{G([4,5]) = [1,2].}$$

There is a topological step hidden in the final interval. A continuous image of a compact interval is compact and connected, so once the extremal values are known the entire range between them is attained. On a noncompact or punctured domain, endpoint values might be approached without being attained.

§1.3 Absolute values are shadows of switching branches

Knowing a range requires knowing where a function changes its formula. An especially useful example is the upper envelope

$$M(x) = \max\{(x-1)^2, (x+1)^2\}.$$

Subtract the candidates: $(x+1)^2 - (x-1)^2 = 4x$. Thus the second branch is active for $x \geq 0$, the first for $x \leq 0$, and

$$\boxed{M(x) = x^2 + 1 + 2|x|.}$$

The identity $\max\{u, v\} = (u + v + |u - v|)/2$ is a reliable way to derive such a formula, with no guesswork about the active branch.

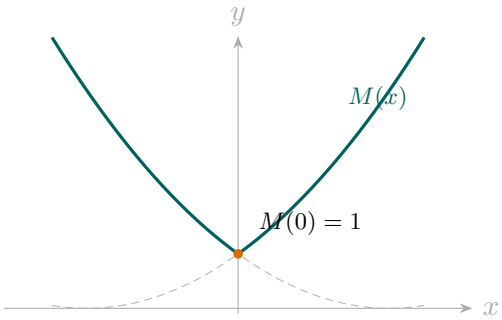


Figure 1.1: The solid upper envelope switches quadratic branches at $x = 0$. The minimum is a corner, so it cannot be found by solving the ordinary equation $M'(0) = 0$.

For $x < 0$ the derivative is $2x - 2$, and for $x > 0$ it is $2x + 2$. At the minimum $x = 0$, the one-sided derivatives are -2 and 2: the ordinary derivative does not exist. Yet convexity provides a better replacement. A number s is a *subgradient* of M at zero if

$$M(y) \geq M(0) + sy \quad \text{for every } y.$$

Since $M(y) - M(0) = y^2 + 2|y|$, every $s \in [-2, 2]$ satisfies the condition. If $|s| > 2$, a sufficiently small y with the same sign as s violates it. Hence

$$\boxed{\partial M(0) = [-2, 2].}$$

The familiar smooth necessary condition $M'(x_*) = 0$ becomes $0 \in \partial M(x_*)$ for convex nonsmooth functions. Here the subgradient interval describes *all* supporting slopes at the corner.[1]

Switching can also change the number of solutions at one parameter value. The simpler function

$$H(x) = |x-1| + |x+1| = \begin{cases} -2x, & x < -1, \\ 2, & -1 \leq x \leq 1, \\ 2x, & x > 1 \end{cases}$$

has no solutions to $H(x) = c$ for $c < 2$, a whole *interval* of solutions for $c = 2$, and precisely two solutions $x = \pm c/2$ for $c > 2$. A critical value can create an entire plateau, in contrast to the pair of roots created at the smooth tangency in the logarithmic example below.

[1] For maxima of differentiable convex functions and subgradients of active branches, see Boyd and Vandenberghe, *Subgradients*, EE364b notes.

§1.4 Dense Sets and Integrability

Let $D(x) = \mathbf{1}_{\mathbb{Q}}(x)$. Rationals and irrationals are dense in every interval, so D is discontinuous at every point. On any nontrivial closed interval, each subinterval has supremum 1 and infimum 0; its upper and lower Riemann sums differ by the whole interval length. Hence D is nowhere Riemann integrable on a nontrivial interval.

Yet $\mathbb{Q}$ has Lebesgue measure zero, so its Lebesgue integral is zero:

$$\int_a^b D(x)\,dx = 0 \quad \text{in the Lebesgue sense.}$$

This is a structural distinction between two integrals. Riemann sums examine the oscillation on every subinterval, while Lebesgue integration can disregard a measure-zero set.

Thomae's function makes the boundary sharper:

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms, } q > 0, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous exactly at the rational points and continuous at every irrational point. To see continuity at an irrational x_0, fix $\varepsilon > 0$ and choose N with $1/N < \varepsilon$. A bounded neighborhood contains only finitely many rationals with denominator $q \le N$. Shrink the neighborhood to avoid them. All remaining values of T are less than ε. Since the discontinuity set is countable, the Lebesgue criterion for Riemann integrability shows T is Riemann integrable with integral 0.

§1.5 A functional equation is transport along an orbit

Suppose $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x+1) = 1 - f(x).$$

Applying the shift twice gives $f(x+2) = f(x)$. Every x has a unique decomposition $x = t + n$ with $0 \le t < 1$ and $n \in \mathbb{Z}$. If $h = f|_{[0,1)}$, then

$$f(t+n) = \begin{cases} h(t), & n \text{ even}, \\ 1 - h(t), & n \text{ odd}. \end{cases}$$

Thus all solutions are determined by arbitrary data on a fundamental interval, transported by the integer translation action.

Continuity is an additional constraint. If h is continuous on $[0,1)$ and has a limit as $t \uparrow 1$, the extension is continuous on $\mathbb{R}$ precisely when

$$\lim_{t \uparrow 1} h(t) = 1 - h(0).$$

For $h(t) = 2t - t^2$, both sides are 1, so the resulting period-two function is continuous.

Write $g = f - \frac{1}{2}$. Then $g(x+1) = -g(x)$, an *antiperiodicity* relation. If g is square-integrable on one period and expanded in the 2-periodic Fourier basis $e^{\pi i k x}$, the shift multiplies the k-th mode by $(-1)^k$. Antiperiodicity therefore forces the even Fourier coefficients to vanish. The alternating-value functional equation has become a spectral selection rule.

§1.6 A quadratic recurrence hides a circle-doubling map

An iteration problem looks quite different from an ordinary range problem, but it becomes manageable when the coordinates expose the repeated operation. Consider

$$x_{n+1} = x_n^2 - 2, \qquad x_0 \in [-2, 2].$$

Expanding $x_2 = (x_0^2 - 2)^2 - 2$ is easy, but expanding x_{10} is a hopeless way to understand its behavior. Instead write $x_0 = 2\cos\theta$, which is possible for every $x_0 \in [-2, 2]$. The double-angle identity gives

$$(2\cos\theta)^2 - 2 = 4\cos^2\theta - 2 = 2\cos(2\theta).$$

Applying the same calculation repeatedly proves by induction that

$$\boxed{x_n = 2\cos(2^n\theta).}$$

The quadratic recurrence has become the doubling of an angle on the circle. This is a *semiconjugacy*

$$h(2\theta) = F(h(\theta)), \qquad h(\theta) = 2\cos\theta, \quad F(x) = x^2 - 2.$$

It is not a conjugacy: θ and $-\theta$ generally represent distinct angles but have the same image under h.

The trigonometric coordinate gives exact periodic orbits. Take

$$x_0 = 2\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{2}.$$

Then

$$x_1 = 2\cos\frac{4\pi}{5} = -\frac{\sqrt{5}+1}{2}, \qquad x_2 = 2\cos\frac{8\pi}{5} = x_0.$$

This is an exact two-cycle. The angle is multiplied by two each time, but taking the cosine forgets reflection of the angle; that is precisely why $8\pi/5$ and $2\pi/5$ give the same x.

The interval $[-2, 2]$ is invariant because the cosine formula always stays inside it. Outside it, the behavior is completely different. If $x_0 > 2$, then

$$x_{n+1} - x_n = (x_n - 2)(x_n + 1) > 0;$$

a bounded increasing orbit would converge to a fixed point of F, but its only fixed points are 2 and -1. Since every iterate is greater than two, the orbit must diverge to $+\infty$. If $x_0 < -2$, the first iterate exceeds 2, and the same conclusion follows.

This is a good example of a natural substitution doing more work than brute force. The algebraic recurrence, the geometry of the circle, and the distinction between bounded and escaping orbits all become explicit from a single identity. The relation with the doubling map is a standard example in one-dimensional dynamics; compare the discussion of the Chebyshev quadratic in *The Doubling Map and Chaos*.

§1.7 Solving a reciprocal functional equation completely

Suppose on $\mathbb{R} \setminus \{0\}$ a function satisfies

$$f(x) = f(1)x + \frac{f(2)}{x} - 1.$$

Set $A = f(1)$, $B = f(2)$. Substituting $x = 1$ gives $B = 1$; substituting $x = 2$ then gives

$$1 = 2A + \frac{1}{2} - 1, \qquad A = \frac{3}{4}.$$

Consequently the function is uniquely determined:

$$\boxed{f(x) = \frac{3x}{4} + \frac{1}{x} - 1.}$$

On $x > 0$, AM–GM gives

$$f(x) \geq 2\sqrt{\frac{3}{4}} - 1 = \sqrt{3} - 1,$$

with equality when $3x/4 = 1/x$, at $x = 2/\sqrt{3}$. A relation that initially seems to involve two unknown function values has reduced to two algebraic consistency equations.

§1.8 Tangency Changes Solution Counts

For $a > 0$, consider

$$\log(ax) = 2\log(x + 1).$$

The real domain is $x > 0$. Exponentiating yields

$$ax = (x + 1)^2 \quad \Longleftrightarrow \quad a = x + 2 + \frac{1}{x}.$$

The last expression is at least 4, with equality only at $x = 1$. Therefore there are no solutions when $0 < a < 4$, exactly one when $a = 4$, and two when $a > 4$.

The same conclusion arises from the discriminant of $x^2 + (2 - a)x + 1$:

$$\Delta = a(a - 4).$$

But the function $a(x) = x + 2 + 1/x$ adds geometric information: the two branches coalesce at the nondegenerate minimum $x = 1$. Near the critical point,

$$a(x) = 4 + (x - 1)^2 + O((x - 1)^3),$$

so if $a = 4 + \varepsilon$ with $\varepsilon > 0$ small, the two solutions satisfy $x = 1 \pm \sqrt{\varepsilon} + O(\varepsilon)$. A discriminant-zero event is a simple fold bifurcation in the geometry of the zero set.

Domains and ranges determine where a function exists and which values it attains; the density of exceptional sets determines regularity and integrability; group actions transport functional equations; and critical values explain how solutions appear and disappear. Each viewpoint becomes useful exactly when the corresponding calculation demands it.